

Ex. 3. $13 \times 252 \div 42 + 170 = ? + 47$

(Bank P.O., 2010)

Sol. Let $13 \times 252 \div 42 + 170 = x + 47$. Then,

$$13 \times \frac{252}{42} + 170 = x + 47 \Rightarrow 13 \times 6 + 170 = x + 47 \Rightarrow x + 47 = 78 + 170 = 248$$

$$\Rightarrow x = 248 - 47 = 201.$$

Hence, required number = 201.

Ex. 4. Simplify: (a) $460 \times 15 - 5 \times 20$

(Bank P.O., 2010)

(b) $1 \div [1 + 1 \div \{1 + 1 \div 1(1 + 1 \div 2)\}] + 1$

(A.A.O., 2009)

Sol. (a) Given expression = $6900 - 100 = 6800$.

$$\begin{aligned} \text{(b) Given expression} &= 1 \div \left[1 + 1 \div \left\{ 1 + 1 \div 1 \left(1 + \frac{1}{2} \right) \right\} \right] + 1 \\ &= 1 \div \left[1 + 1 \div \left\{ 1 + 1 \div \frac{3}{2} \right\} \right] + 1 = 1 \div \left[1 + 1 \div \left\{ 1 + 1 \times \frac{2}{3} \right\} \right] + 1 \\ &= 1 \div \left[1 + 1 \div \left\{ 1 + \frac{2}{3} \right\} \right] + 1 = 1 \div \left[1 + 1 \div \frac{5}{3} \right] + 1 \\ &= 1 \div \left[1 + 1 \times \frac{3}{5} \right] + 1 = 1 \div \left[1 + \frac{3}{5} \right] + 1 = 1 \div \frac{8}{5} + 1 = 1 \times \frac{5}{8} + 1 = \frac{5}{8} + 1 = \frac{13}{8}. \end{aligned}$$

Ex. 5. Find the missing numeral:

(a) $(? - 2763) \div 86 \times 13 = 208$

(L.I.C.A.D.O., 2007)

(b) $3565 \div 23 + 4675 \div ? = 430$

(Bank Recruitment, 2008)

Sol. (a) Let $(x - 2763) \div 86 \times 13 = 208$.

$$\text{Then, } \frac{(x - 2763)}{86} \times 13 = 208 \Rightarrow \frac{(x - 2763)}{86} = \frac{208}{13} = 16$$

$$\Rightarrow (x - 2763) = 16 \times 86 = 1376$$

$$\Rightarrow x = 1376 + 2763 = 4139.$$

(b) Let $3565 \div 23 + 4675 \div x = 430$.

$$\text{Then, } \frac{3565}{23} + \frac{4675}{x} = 430 \Rightarrow 155 + \frac{4675}{x} = 430$$

$$\Rightarrow \frac{4675}{x} = 430 - 155 = 275 \Rightarrow x = \frac{4675}{275} = 17.$$

Ex. 6. Simplify: (a) $\frac{(6 + 6 + 6 + 6) \div 6}{4 + 4 + 4 + 4 + 4}$ (b) $\frac{(2 + 3) \times 5 + 3 \div \frac{1}{2}}{6 + 5 \times 4 \div \frac{4}{5}}$

(R.R.B., 2006)

Sol. (a) Given expression = $\frac{24 \div 6}{4 + 4 + 4 + 1} = \frac{4}{13}$.

$$\text{(b) Given expression} = \frac{5 \times 5 + 3 \times 2}{6 + 5 \times 4 \times \frac{5}{4}} = \frac{25 + 6}{6 + 25} = \frac{31}{31} = 1.$$

Ex. 7. What should come in place of both the question marks in the following equation?

$$\frac{128 + 16 \times ? - 7 \times 2}{7^2 - 8 \times 6 + ?^2} = 1$$

Sol. Let $\frac{128 + 16 \times x - 7 \times 2}{7^2 - 8 \times 6 + x^2} = 1$.

$$\text{Then, } 8x - 7 \times 2 = 49 - 48 + x^2 \Leftrightarrow 8x - 14 = 1 + x^2 \Leftrightarrow x^2 - 8x + 15 = 0$$

$$\Leftrightarrow x^2 - 3x - 5x + 15 = 0 \Leftrightarrow x(x - 3) - 5(x - 3) = 0$$

$$\Leftrightarrow (x - 3)(x - 5) = 0 \Leftrightarrow x = 3 \text{ or } x = 5.$$

Hence, missing number is 3 or 5.